Generell Kjemi øving 1 uke 37

ØVING 1

1a) $n = 1{,}38\,mol$ $V = 5{,}21\,dm^3$ $T = 648\,K$ $p = ?$ $pV = nRT$

$R = 8{,}314\,J\,mol^{-1}K^{-1}$ $p = \frac{nRT}{V}$

$$p = \frac{1{,}38\,mol \cdot 8{,}314\,J\,mol^{-1}K^{-1} \cdot 648\,K}{5{,}2 \cdot 10^{-3}\,m^3} = \frac{7434{,}711\,J}{5{,}2 \cdot 10^{-3}\,m^3} = \frac{7434{,}711\,Nm}{5{,}2 \cdot 10^{-3}\,m^3}$$

$p = 1429752{,}115\,N/m^2 = \underline{\underline{14{,}3 \cdot 10^5\,Pa}}$

b) $n = 1{,}3\,mol$ $V = 74{,}1\,dm^3$ $p = 45\,atm$ $T?$ $T = \frac{pV}{nR}$

$p = 45 \cdot 101{,}3 \cdot 10^3 = 4558500\,Pa$

$$T = \frac{4558500\,pa \cdot 74{,}1 \cdot 10^{-3}\,m^3}{1{,}3\,mol \cdot 8{,}314\,J\,mol^{-1}K^{-1}} = \frac{4558500\,Nm^{-2} \cdot 74{,}1 \cdot 10^{-3}\,m^3}{1{,}3 \cdot 8{,}314\,Nm\,K^{-1}}$$

$T = 31252{,}65\,K = \underline{\underline{3{,}1 \cdot 10^4\,K}}$

c) $\rho = ?$ $T = 298\,K$ $p = 10{,}0\,atm$ $M_m = 18\,g/mol$ $\rho = \frac{m}{V}$ $pV = nRT$ $n = \frac{m}{M_m}$

$pV = \left(\frac{m}{M_m}\right)RT \rightarrow \frac{m}{V} = \frac{pM_m}{RT} \rightarrow \rho = \frac{pM_m}{RT}$ $p = 10{,}0 \cdot 101{,}3 \cdot 10^3 = 101{,}3 \cdot 10^4$

$$\rho = \frac{101{,}3 \cdot 10^4\,Pa \cdot 18\,g/mol}{8{,}314\,J\,mol^{-1}K^{-1} \cdot 298\,K} = \frac{101{,}3 \cdot 10^4\,N/m^2 \cdot 18\,g}{8{,}314\,Nm \cdot 298} = 7359{,}62\,gm^{-3}$$

$\underline{\underline{\rho = 7{,}4\,g/L}}$

2) $2Al + 6HCl \rightarrow 3H_{2(g)} + 2AlCl_3$

$m = 2{,}35\,g$

$M_m = 26{,}98\,g/mol$

$n = 0{,}0871\,mol \xrightarrow{2:3} n = 0{,}13065\,mol$

$P_H = P_{tot} - P_{vannsøle} = (754 - 17{,}53)\,mm\,Hg = 736{,}47\,mm\,Hg$

$T = 293\,K$

$$V = \frac{nRT}{p} = \frac{0{,}131\,mol \cdot 8{,}314\,Nm\,mol^{-1}K^{-1} \cdot 293\,K}{(736{,}47 \cdot 133{,}3)\,N/m^2}$$

$V = 3{,}25\,m^3 = \underline{\underline{3{,}25\,L}}$

ØVING 1

3) a) $\rho = ?$ $M_m = 28{,}96\,g/mol$ $p = 1\,atm = 1{,}013 \cdot 10^5\,Pa$ $T = 298\,K$

$pV = nRT \rightarrow pV = \frac{m}{M_m}RT \rightarrow \frac{m}{V} = \frac{pM_m}{RT} \rightarrow \rho = \frac{pM_m}{RT}$

$$\rho = \frac{1{,}013 \cdot 10^5\,N/m^2 \cdot 28{,}96\,g/mol}{8{,}314\,Nm\,mol^{-1}K^{-1} \cdot 298\,K} = 1184{,}08\,g/m^3 = \underline{1{,}18\,g/L}$$

b) $M_{m_{CO_2}} = 44{,}01$ $p = 1\,atm = 1{,}013 \cdot 10^5\,Pa$ $T = 298\,K$

$$\rho = \frac{1{,}013 \cdot 10^5\,N/m^2 \cdot 44{,}01\,g/mol}{8{,}314\,Nm\,mol^{-1}K^{-1} \cdot 298\,K} = 1799{,}43\,g/m^3$$

ρ_{CO_2} er høyere enn ρ_{luft}, ballonger med CO_2 vil derfor synke

4) $Cl_{2(g)} + CO_{(g)} \rightarrow COCl_{2(g)}$

$n = 2\,mol$ $n = 1\,mol$

CO er den begrensende reaktanten

$p = ?$ $T = 375\,K$ $n = 1\,mol$ $V = 1\,m^3$

$$p = \frac{nRT}{V} = \frac{2\,mol\ 8{,}314\,Nm\,mol^{-1}K^{-1} \cdot 375\,K}{1\,m^3} = 6235{,}5\,Pa = \underline{6{,}24\,kPa}$$

fordi det er 2 mol med stoffer i reaktanten

5) $U = \frac{3}{2}nRT$ $[U] = J$

a) $U = \frac{3}{2} \cdot 1\,mol\ 8{,}314\,J\,mol^{-1}K^{-1} \cdot 345\,K = 4302{,}495 = \underline{4{,}3\,kJ}$

b) $U = \frac{3}{2} \cdot 2\,mol\ 8{,}314\,J\,mol^{-1}K^{-1} \cdot 298\,K = 7432{,}716\,J = \underline{7{,}4\,kJ}$

c) $U = \frac{3}{2} \cdot 1\,mol \cdot 8{,}314\,J\,mol^{-1}K^{-1} \cdot 200\,K = 2494{,}2\,J = \underline{2{,}5\,kJ}$

d) $U = \frac{3}{2} \cdot 1\,mol \cdot 8{,}314\,J\,mol^{-1}K^{-1} \cdot 6\,K = 74{,}826\,J = \underline{0{,}075\,kJ}$

ØVING 1

6) $M_{\%,C} = 46,2$ $M_{\%,N} = 53,8$ $T = 298K$ $p = 750\ torr = 99975\ pa$

$m = 1g$ $V = 0,476L = 0,476 \cdot 10^{-3} m^3$ $pV = nRT$

$n = pV/RT$

$$n = \frac{99975 \cancel{N/m^2} \cdot 0,476 \cdot 10^{-3} \cancel{m^3}}{8,314 \cancel{Nm}\ mol^{-1} \cancel{K^{-1}} \cdot 298 \cancel{K}} = 0,0192\ mol = 1,92 \cdot 10^{-2}\ mol$$

$$M_{MCN} = m/n = \frac{1g}{1,92 \cdot 10^{-2}\ mol} = 52,08$$

$M_{mC} = 12,01\ g/mol$ $M_{mN} = 14,01\ g/mol$

finner ut av hvor mye det må være av hvert stoff (C&N) i

$$\frac{12,01 \cdot x}{52,08} = 0,462 \Rightarrow x = \frac{0,462 \cdot 52,08}{12,01} = 2,0034 \approx 2$$

$$\frac{14,01 \cdot x}{52,08} = 0,538 \Rightarrow x = \frac{0,538 \cdot 52,08}{14,01} = 1,9999 \approx 2$$

C_2N_2